

Radio brief: band, module and link budget

Team Santiago, Andrian, Marcos **Version** 0.2 · Sun 4 Oct 2026 **Next session** Fri 9 Oct (week 3)

New in v0.2: a working baseline (868 MHz + RFM69HCW), the duty-cycle limit that now drives our planning, the electronics handoff, a shared-PHY proposal for the other teams, and the test plan.

What the rules allow, what the numbers say, and what we need to decide first. Every figure carries a tag: **RULE** from the competition rules PDF, **VERIFIED** against a primary source, **ASSUMPTION** to be replaced by a measurement, and **ESTIMATE** .

Bottom line

1. **Only two bands allow a 2-minute continuous round:** 869.40–869.65 MHz (500 mW e.r.p., $\leq 10\%$ duty cycle) and 2.4 GHz (10 mW e.i.r.p.). The 433 MHz antenna does not fit the box and the other 868 sub-bands fail on duty cycle. VERIFIED
2. **868 MHz has range to spare but limited speed.** Margin is +36 to +44 dB at 70 m and +19 to +21 dB at 260 m. The 250 kHz channel caps the air rate at about 100–150 kbps. That just covers our goal of 10,000 lines in 2 minutes (it needs ~67 kbps of useful throughput), so compression gives us the safety margin.
3. **2.4 GHz (nRF24L01+) has speed to spare but thin margin:** +6 to +9 dB at 70 m and –4.6 dB at 260 m with small whips. Directional 8 dBi patches bring 260 m back to +7 dB.
4. **Our band choice depends on the other teams.** In NM every team must share one physical layer, and in MRM every team transmits at once. So the band talks with the other teams have to start *this week*, before we commit to a module. The module is frozen at Quick Mode, and changing it afterwards costs every bonus. RULE
5. **Working baseline (v0.2):** 868 MHz at 869.525 MHz with the **RFM69HCW** at +17 dBm and 100 kbps. 2.4 GHz nRF24 stays an add-on option for MRM. Santiago adopted this on 4 Oct so build work can start. **Andrian and Marcos still need to sign off, by Fri 16 Oct at the latest.**
6. **Duty cycle, not range or speed, is the binding limit.** A 2-minute sending round is about 115 s of TX, and each box may transmit 360 s per hour. That is **about 3 full rounds per hour per box, campus tests included.** Repeated tests therefore go to 869.70–870.00 MHz (5 mW, no duty-cycle limit).

Working baseline sign-off pending

We chose the RFM69HCW over the Si4463 because range is not our problem at 868 MHz. It is easier to bring up (mature open drivers, simple registers), other teams are more likely to pick it, which helps the shared NM link, and it carries less protocol-ban risk (the Si4463 advertises 802.15.4g). We run it at **+17 dBm**, because the datasheet limits +20 dBm to 1 % duty cycle.

VERIFIED

Four radio profiles are defined. Each is checked against the CNAF sub-band (Carson bandwidth plus ± 20 ppm crystal drift per box ASSUMPTION) and the SX1231 datasheet limits.

PROFILE	USE	CARRIER	RATE	DEVIATION	PA	E.R.P.	FITS BAND?
SRI_100K	Our own link: SRI,	869.525 MHz	100 kbps	± 50 kHz	+17 dBm	27 mW	yes · 7.6 kHz gr

PROFILE	USE	CARRIER	RATE	DEVIATION	PA	E.R.P.	FITS BAND?
	QM, MRM						
ROBUST_38K4	Fallback if 100 kbps fails (+5 dB)	869.525 MHz	38.4 kbps	±38.4 kHz	+17 dBm	27 mW	yes · 50 kHz gu
NM_COMMON	Shared with all teams (draft, below)	869.525 MHz	100 kbps	±50 kHz	+17 dBm	27 mW	yes · 7.6 kHz g
BENCH_5MW	Bench and repeated tests, no duty limit	869.850 MHz	100 kbps	±50 kHz	+9 dBm	4 mW	yes · 32.6 kHz

Duty-cycle budget per box

The model uses selective-repeat ARQ with one ACK per 16 frames.

- Sender TX per 2-min round: **115.5 s**
- Receiver TX (ACKs) per round: **2.0 s**
- Budget in any one hour: **360 s** (10 %)
- Full sending rounds per hour: **≈3.1**

The firmware refuses to transmit past 95 % of the hourly budget.

Firmware ready before the hardware

A portable C driver for the RFM69 is written and passes 154 host tests against a simulated radio. It needs only five board functions (SPI select and transfer, reset pin, delay, millisecond clock), so it fits whichever micro the electronics team picks.

It includes the duty-cycle guard and the software CRC for the NM link. Not yet run on a real module.

First antenna (QM)

A $\lambda/4$ wire (86 mm) bent as an inverted-L: about 60 mm up the box wall, 25 mm under the lid. Cut it long and trim for best RSSI.

Mount the radio and antenna **as high in the box as possible**. Also buy one 868 MHz FPC or

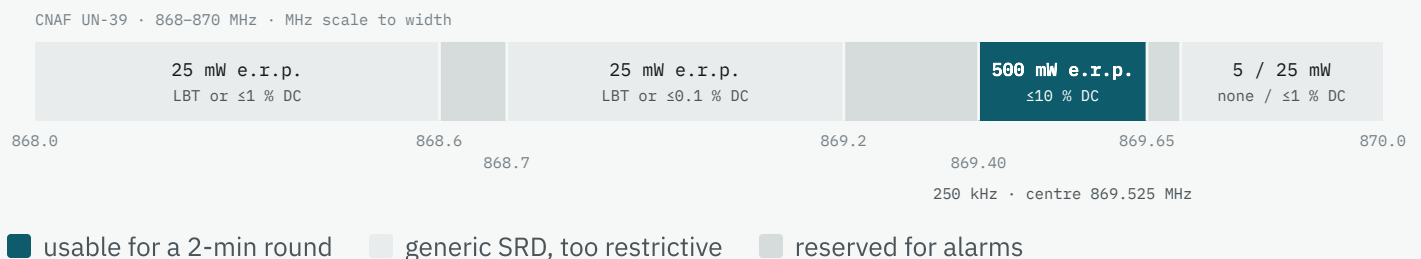
helical antenna for an A/B test. Final antenna by week 8, after the antenna consultants review it.

Rules that constrain the radio

- Unlicensed spectrum only, compliant with **CNAF 2017 UN notes**, IEEE C95.1 (RF exposure) and ESD safety. **RULE**
- **No chip implementing a standard wireless access protocol** (WiFi, Bluetooth, 2G–5G, ZigBee; the Session 1 slide adds NB-IoT and **LoRa**). **RULE**
- **Nothing may stick out of the 15 × 15 × 7 cm IKEA 365+ box** except buttons, switches and connectors, so the antennas go inside. The box sits on a support of **about 70 cm**. **RULE**
- **Modules shown at Quick Mode are final**. Antennas, batteries and mechanics may still change. **RULE**
- Distances: SRI **70 m**, MRM **~260 m** with every team transmitting at once, NM relays along the South and East rims with no line of sight end to end. The link must recover from "attacks". **RULE**
- Scoring counts only the **first unbroken run of byte-identical lines** (~10,000 lines of ~100 B), so reliability beats raw bit rate. **RULE**

Spectrum: what CNAF allows

I checked the BOE text of CNAF 2017 (the edition the rules cite) against the current CNAF 2021; the limits we care about are the same in both. Duty cycle is measured over any one hour (ERC/REC 70-03). **VERIFIED**



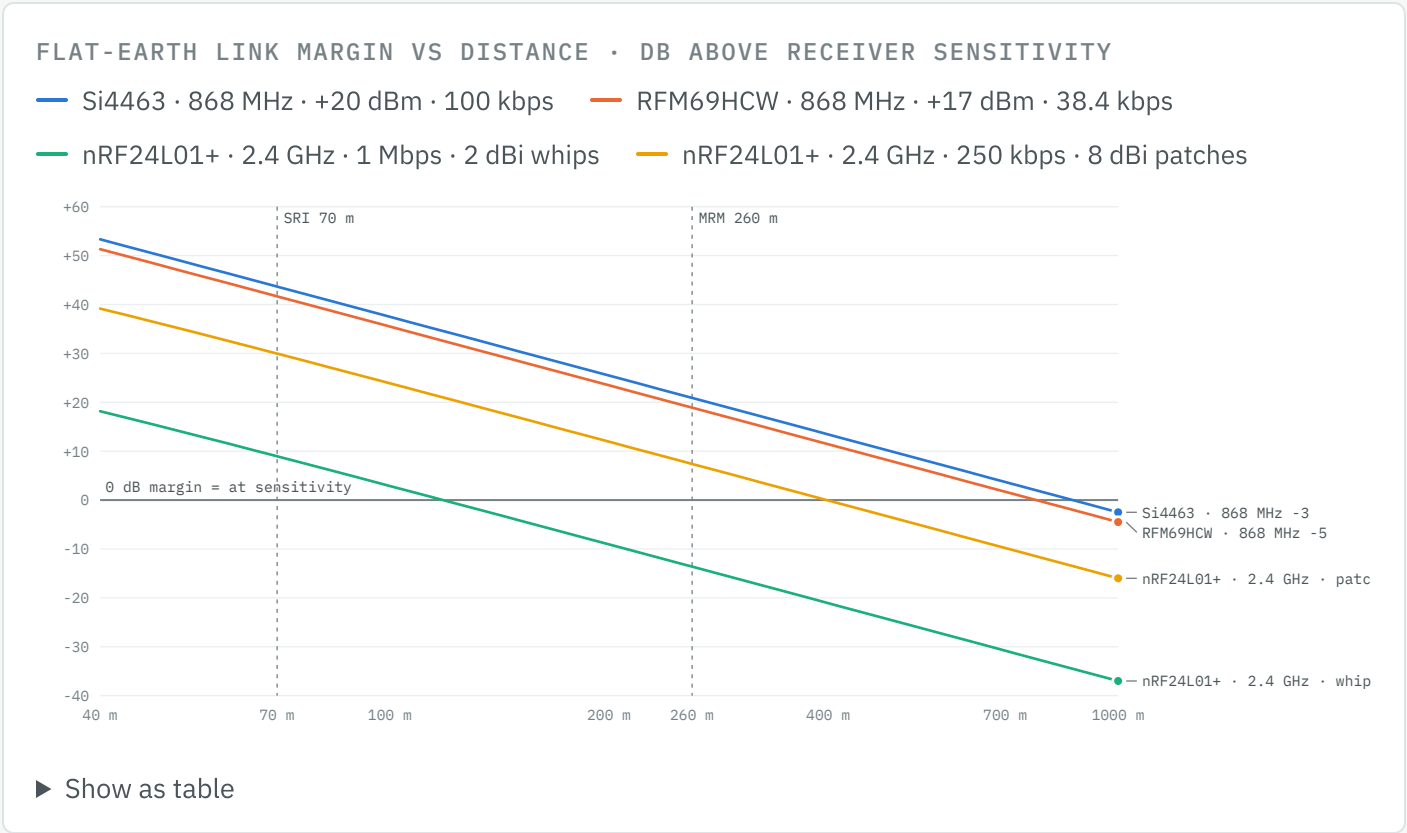
BAND	NOTE	MAX POWER	ACCESS CONDITION	2-MIN ROUND?	COMMENT
433.05–434.04 MHz	UN-30	10 mW e.r.p.	≤10 % DC	legal, impractical	$\lambda/4 = 17$ cm, longer than the box.
868.0–868.6 MHz	UN-39	25 mW e.r.p.	LBT+AFA or ≤1 % DC	no	1 % is 36 s per hour.
869.40–869.65 MHz	UN-39	500 mW e.r.p.	LBT+AFA or ≤10 % DC	yes	360 s per hour. SRI 120 s + MRM 120 s = 240 s, even if the same box transmits both. The whole 250 kHz may be one high-speed channel.
869.70–870.00 MHz	UN-39	5 mW e.r.p.	none	yes, 5 mW	Weak, but a possible low-power NM fallback.
2400–2483.5 MHz	UN-85 b)	10 mW e.i.r.p.	none (EN 300 440)	yes	83.5 MHz wide: room for every team on its own channel in MRM. Shared with campus WiFi.
2400–2483.5 MHz	UN-85 a)	100 mW e.i.r.p.	mitigation required (EN 300 328)	ask Atenea	+10 dB, but it is the WiFi/RLAN class. Whether a student FHSS or LBT design counts needs an official answer.
5725–5875 MHz	UN-130	25 mW e.i.r.p.	none	legal, impractical	Few cheap non-WiFi modules on the market.

Sources: [CNAF 2017](#), [BOE-A-2017-12318](#) · [CNAF 2021](#), [BOE-A-2021-21346](#). In Spain 915 MHz is not an SRD band, so US "915 MHz" module variants are not usable.

Link budget

This uses the course method: Maxim AN5142 with the course correction (no ½ factor in the flat-earth formula) and the course spreadsheet's worst-case ground bounce (a = 1, |cos|). Inputs:

- antennas at 0.75 m (70 cm support plus position in the box) **ASSUMPTION**
- 1 dB container loss and 0 dB multipath on open campus paths **ASSUMPTION**
- 868 MHz antenna gain 0 dBi in the box **ASSUMPTION**
- sensitivities from the datasheets **VERIFIED**



SCENARIO	EIRP DBM	RX DBM	SENS. DBM	MARGIN DB	WITH -20 DB MULTIPATH
SRI 70 m RFM69HCW 868 38.4 kbps	16.5	-63.3	-105.0	+41.7	+21.7
SRI 70 m RFM69HCW 868 100 kbps	16.5	-63.3	-100.0	+36.7	+16.7

SCENARIO	EIRP DBM	RX DBM	SENS. DBM	MARGIN DB	WITH -20 DB MULTIPATH
SRI 70 m Si4463 868 100 kbps	19.5	-60.3	-104.0	+43.7	+23.7
SRI 70 m CC1101 868 38.4 kbps	11.5	-68.3	-104.0	+35.7	+15.7
SRI 70 m nRF24 2.4G 1 Mbps	2.0	-76.0	-85.0	+8.9	-11.1
SRI 70 m nRF24 2.4G 2 Mbps	2.0	-76.0	-82.0	+6.0	-14.1
MRM 260 m RFM69HCW 868 38.4 kbps	16.5	-86.1	-105.0	+18.9	-1.1
MRM 260 m Si4463 868 100 kbps	19.5	-83.1	-104.0	+20.9	+0.9
MRM 260 m nRF24 2.4G 250 kbps	2.0	-98.6	-94.0	-4.6	-24.6
MRM 260 m nRF24 2.4G 250 kbps, 8 dBi patches	8.0	-86.6	-94.0	+7.4	-12.6
NM hop 100 m RFM69HCW 868 38.4 kbps, -10 dB obstr.	16.5	-79.5	-105.0	+25.5	+5.5
NM hop 100 m nRF24 2.4G 250 kbps, -10 dB obstr.	2.0	-92.1	-94.0	+1.9	-18.1

Full AN5142 rows for all 12 scenarios are in `radio/MTP-F26_TeamC_Radio_LinkBudget.xlsx`; every cell is a live formula and inputs are in blue. AN5142's default -20 dB multipath is for cluttered paths. Tests T4 and T5 on campus will tell us which number to use.

Antenna height is the biggest lever

At 0.75 m the ground reflection takes over beyond ~13 m (868) or ~37 m (2.4 GHz). Past that point loss grows 40 dB per decade and hardly depends on frequency: $P_R \approx \text{EIRP} \cdot G_R \cdot (h_1 h_2)^2 / d^4$.

Raising both antennas from 0.5 to 1.0 m adds **+12 dB**. Mounting the antenna at the top of the

The 868 MHz antenna is longer than the box is tall

A 868 MHz quarter-wave is ~86 mm; the box is 70 mm tall. Our options are a helical or meander antenna (shorter, lower gain), a bent inverted-L, a PCB antenna, or a horizontal wire or dipole laid across the 21 cm diagonal.

A horizontal antenna gives up the all-round coverage NM needs. At 2.4 GHz a vertical $\lambda/4$

box is free margin. Whether "about 70 cm" lets us use a taller support is a question for Atenea.

is 31 mm and fits easily. Even at -5 dBi per end, 868 keeps +30 dB at 70 m and +9 dB at 260 m.

Datasheet limits that matter

- **RFM69HCW**: +20 dBm is limited to **1 % duty cycle** (datasheet §3.3.7), so plan on +17 dBm for 2-minute rounds. **VERIFIED**
- **CC1101 at 250 kBaud** needs a 540 kHz filter, wider than the 250 kHz sub-band. Its practical ceiling there is ~100 kbps.
- **nRF24L01+**: 32 B maximum payload, 0 dBm. PA+LNA variants must be turned down to stay within 10 mW e.i.r.p.

Throughput: does the file fit in 2 minutes?

About 1 MB in 120 s needs **66.7 kbps** of useful throughput, or 33.3 kbps with 2× compression. The model assumes selective-repeat ARQ with one ACK per 16 frames, 1 ms turnaround **ASSUMPTION**, and 85 % efficiency for the micro and USB **ASSUMPTION**. Bandwidth uses Carson's rule **ESTIMATE**.

CONFIGURATION	AIR RATE	GOODPUT	LINES, RAW	LINES, 2× COMP.	FITS 250 KHZ?
868 FSK, 60 B payload	38.4 kbps	26.2 kbps	3,924	7,849	yes · ~77 kHz
868 FSK, 60 B payload	100 kbps	67.3 kbps	10,000	10,000	yes · ~200 kHz
868 GFSK h = 0.5, 60 B payload	150 kbps	99.9 kbps	10,000	10,000	tight · ~225 kHz
868 shared in MRM (5 × 50 kHz)	19.2 kbps	13.1 kbps	1,970	3,940	yes · ~38 kHz each
nRF24 2.4 GHz, 32 B payload	250 kbps	158.7 kbps	10,000	10,000	n/a
nRF24 2.4 GHz, 32 B payload	1 Mbps	594.1 kbps	10,000	10,000	n/a

At 868 MHz the whole file only arrives at ≥ 100 kbps. Sensitivity then drops about 5 dB ($-105 \rightarrow \sim -100$ dBm on RFM69 ESTIMATE), but at 70 m we still keep more than +35 dB. In MRM it is different: if every team shares 869.4–869.65 MHz, each gets roughly 2,000–4,000 lines. That is why the MRM band is a negotiation with the other teams.

Module shortlist

Figures were read from each datasheet on 3 Oct 2026. Prices are still **TO VERIFY**: the electronics team should get distributor quotes, because cost only pays out among NM winners.

MODULE	BAND	MAX TX	SENSITIVITY	MAX
Si4463 (RFM26W and similar)	142–1050 MHz	+20 dBm · 85 mA	–109 @40k · –104 @100k · –97 @500k	500
RFM69HCW (SX1231H) baseline	868 variant	+17 dBm cont. · 95 mA	–114 @4.8k · –105 @38.4k	300

MODULE	BAND	MAX TX	SENSITIVITY	MAX
CC1101	779– 928 MHz	+12 dBm · 34 mA	–104 @38.4k · –95 @250k	500
nRF24L01+	2.4 GHz	0 dBm · 11.3 mA	–94 @250k · –85 @1M · –82 @2M	2 Mb
CC2500	2.4 GHz	+1 dBm	–89 @250k · –83 @500k	500
LoRa chips (SX127x/RFM95, SX126x, SX1280)	—	—		
Radio SoCs (CC1310, nRF52, ESP32, STM32WL)	—	—		

Decisions and who they touch

DECISION	OWNER	DEPENDS ON	AFFECTS	STATUS
D-R1 Primary band (868 vs 2.4 GHz vs both)	Radio	Other teams' NM/MRM band intentions; Atenea answers	Module, antennas, firmware	baseline: 868 MHz sign-off by Fri 16 Oct
D-R2 Radio module (frozen at Quick Mode)	Radio → handed to electronics	D-R1; bench sensitivity at ≥100 kbps	PCB, micro interface (SPI), BoM	baseline: RFM69HCW sign-off by Fri 16 Oct; gate: test T2
D-R3 NM physical layer shared with all teams (frequency, rate, modulation, sync word, CRC)	Radio + protocols	What modules the other teams buy	NM bonus (+2 to +2.5)	draft ready send to teams; freeze by week 6
D-R4 MRM access plan (own band, shared channel split, or 2.4 GHz channels)	Radio	Number of teams and their bands; tests at 260 m	MRM (+1)	open week 8–9
D-R5 Antenna per round (in-box gain, polarisation, mounting height)	Radio	Container measurements (open vs closed)	Mechanics, drop test	QM: inverted-L wire final by week 8

Reverse D-R2 if test T2 shows worse than –95 dBm sensitivity at 100 kbps, or Atenea bans the chip. The fallback is the Si4463: same band and link budget, more bring-up effort. **Reverse D-R1** if the other teams converge on 2.4 GHz for NM, or the competition schedule cannot fit the 10 % budget.

Handoff to electronics

The radio team picks the transceiver and the electronics team adapts and picks the micro. This is what the board must provide. Part: **HopeRF RFM69HCW, 868 MHz version** (the 433 and 915 versions have different matching). The bare module is a small SMD part. For QM a ready-made breakout is easier, e.g. Adafruit "RFM69HCW 868/915 MHz" with regulator and level shifting.

Package size and part number are **to verify** against the datasheet and distributor before ordering. Order 4 (2 boxes + 2 spares) from a distributor, not a marketplace.

PIN	AT THE MICRO	WHAT MATTERS
3.3V	supply	3.3 V only, not 5 V tolerant. Budget ≥ 120 mA peak (≈ 95 mA TX at +17 dBm). 10 μ F + 100 nF next to the pin.
SCK MOSI MISO	SPI master	Mode 0, MSB first, ≤ 10 MHz. 4–8 MHz is plenty: 100 kbps is only 12.5 kB/s.
NSS	GPIO out	Chip select, active low.
RESET	GPIO out	Active high. Held low in normal use; the driver pulses it at start-up.
DI00	GPIO in, IRQ-capable	Packet sent (TX) / packet ready (RX).
DI01	optional	Only for frames longer than 64 B.
ANT	—	50 Ω to a short wire or u.FL. Keep it away from the micro, battery and USB.

Also needed on the board: no WiFi/BT chip, and no micro with a built-in radio (protocol ban). What we need back from electronics: the micro and pin map, a power check that the battery and regulator hold the 120 mA TX peak, and module lead time so T2 runs before Quick Mode.

Shared NM radio settings: draft for all teams

In Network Mode every box must hear every other team's box. This is our opening proposal. It works on the three popular sub-GHz chips (SX1231/RFM69, CC1101, Si4463), so teams can still buy hardware that supports it.

SETTING	VALUE	WHY
Carrier	869.525 MHz	Only 868 sub-band with 500 mW and 10 % duty cycle
Modulation · rate · deviation	2-GFSK BT 0.5 · 100 kbps · ± 50 kHz	Fits 250 kHz with drift; all three chips support it
Max TX power	$\leq +17$ dBm conducted	Every box stays well under 500 mW e.r.p.

SETTING	VALUE	WHY
Preamble · sync word	4 × 0xAA · 0x2D 0xD4	That sync word is the CC1101 and RadioHead default
Length	1 byte, variable, max 64	Same convention on all three chips; fits a 64 B FIFO
Whitening · Manchester	off	The chips' whitening isn't guaranteed compatible
CRC	hardware off; CRC-16/CCITT-FALSE in software , last 2 bytes, big-endian	Hardware CRCs differ between vendors. Check value 0x29B1 for "123456789"

Fallback for chips that can't do 100 kbps: 38.4 kbps, ±38.4 kHz, same framing. Interop test before week 6: every pair of boxes exchanges 1 000 frames at 10 m on 869.85 MHz (same settings, outside the duty-limited band); pass is ≥ 99 % with valid CRC. Addressing and relaying sit above this and are for the protocols teams.

Test plan

Use the bench profile (869.85 MHz, 4 mW, no limit) unless a test needs the competition band. Log every TX second there. For each run, record place, profile, distance, heights, box open or closed, frames sent and received, and RSSI.

TEST	WHAT	PASS	WHEN
T0	Bring-up: driver start-up, then 1 000 frames at 1 m	Start-up OK; ≥ 999 frames received	modules arrive
T2	Sensitivity vs bit rate: step TX power and distance down a corridor, 1 000 frames per point	1 % packet errors at RSSI ≤ −95 dBm at 100 kbps. Gate for the module decision.	before Quick Mode
T4	Range, open box, line of sight, at 10 / 70 / 150 / 260 m; then one 2-min transfer at 70 m	RSSI near the model (table below); ≥ 10 000 lines	after T2
T5	Closed box; wire vs bought antenna; antenna at the top vs the bottom of the box	≥ +10 dB margin at 260 m at +17 dBm	after T4

TEST	WHAT	PASS	WHEN
T13	Spectrum analyser: carrier offset, 99 % occupied bandwidth, conducted power	Whole emission inside 869.40–869.65 MHz; e.r.p. $\leq$ 500 mW. Plot goes in the ANNEX.	book the lab

EXPECTED (MODEL)	10 M	70 M	150 M	260 M
Bench profile, +9 dBm · RSSI	−43 dBm	−71 dBm	−85 dBm	−94 dBm
Bench profile · margin	+57 dB	+29 dB	+16 dB	+6 dB
SRI profile, +17 dBm · margin	+65 dB	+37 dB	+24 dB	+14 dB

The difference between measured and expected RSSI is the real multipath and container loss. It replaces the −1 dB / 0 dB assumptions in the workbook.

This week (to Fri 9 Oct)

A suggested split. Swap names as you like; per the course slides, owners volunteer and set their own due dates.

Regulation and outreach

- Read UN-30, UN-39 and UN-85 in the BOE text and sign off the regulatory sheet.
- Draft the Atenea questions below; Marcos posts them as project leader.
- Contact every team's leader: which band and module are you considering? Propose a joint NM physical-layer meeting.

Modules and budget

- Check prices and stock for the 4 modules (Mouser, DigiKey, LCSC, TME).
- Collect each module's register and setup notes for 100 kbps in 250 kHz.
- Update the workbook with any better datasheet numbers.

Antennas and test plan

- Antenna options for 868 and 2.4 inside a 15 × 15 × 7 box: size, gain, polarisation, price.
- Write the test procedures: T2 bench sensitivity vs bit rate, T4/T5 range at 70 m open and closed, T13 spectrum and e.r.p.

- Book Anna Umbert (spectrum regulation) and Concepción Santos / Carles Puente (antennas) for week 3–4 sessions.

Action items in the course format

ITEM	DATE ASG.	ASSIGNED TO	DESCRIPTION	STATUS	DUE DATE	END DATE	RESULT
R1	05/10/26	RT (tbd)	Regulatory sheet UN-30/39/85 verified and signed	Open	09/10/26		
R2	05/10/26	RT (tbd)	Atenea questions drafted and sent to Marcos for posting	Done	07/10/26	04/10/26	Draft ready (below)
R3	05/10/26	RT (tbd)	Band/module survey of all other teams; NM physical-layer meeting proposed	Open	09/10/26		
R4	05/10/26	RT (tbd)	Module price/stock table (4 candidates, 2 distributors each)	Open	09/10/26		
R5	05/10/26	RT (tbd)	Antenna options sheet for 868 and 2.4 inside the box	Open	09/10/26		
R6	05/10/26	RT (tbd)	Consultant sessions booked (AU; CS or CP)	Open	09/10/26		
R7	09/10/26	RT	Band and module decision (D-R1, D-R2)	In progress	16/10/26		Baseline set 04/10; needs RT

ITEM	DATE ASG.	ASSIGNED TO	DESCRIPTION	STATUS	DUE DATE	END DATE	RESULT
			handed to electronics				+ PM sign-off
R8	04/10/26	RT (tbd)	NM radio settings proposal sent to every team's leader	Open	09/10/26		
R9	04/10/26	RT → ET	Electronics handoff reviewed with ET; pin map and power check returned	Open	16/10/26		
R10	04/10/26	ET	Order 4 × RFM69HCW 868 (after sign-off)	Open	16/10/26		
R11	04/10/26	RT	Tests T0 and T2 done; D-R2 confirmed or reversed	Open	before QM		

Questions to ask

Atenea forum (posted by the project leader)

- **Allowed chips.** Is a proprietary FSK transceiver like the RFM69HCW (SX1231H) allowed? It implements no standard protocol. Is a chip that *can* do a standard PHY we never use (e.g. Si4463, 802.15.4g) also acceptable?
- **Duty cycle and the schedule.** 869.40–869.65 MHz allows 10 % of any hour (360 s) per transmitter, and a 2-min sending round uses ~115 s. How will SRI, MRM, NM rounds and rehearsals be scheduled so teams stay within it? Is it assessed per transmitter?

Consultants

- **Anna Umbert** (spectrum regulation): can we rely on the 10 % duty cycle, or should we implement LBT+AFA (EN 300 220)? What e.r.p. evidence does she expect in the ANNEX?
- **Concepción Santos / Carles Puente** (antennas): best compact 868 antenna for a vertically polarised, omnidirectional pattern in a 70 mm-tall plastic box. How does a hand or the foil near the box detune it? Can we measure gain at the lab?
- **Paradells / Perelló** (protocols, with the protocols team): sharing one 250 kHz

- **2.4 GHz power.** May a non-WiFi design with frequency hopping or listen-before-talk use the 100 mW of UN-85 a), or must it stay at 10 mW (UN-85 b)?
- **Support height.** Is the support fixed at 70 cm or "about 70 cm"? May the antenna sit anywhere inside the box, e.g. under the lid?
- **Cooperation.** May teams agree a common NM radio setup and an MRM frequency split, sharing settings but not code?

channel among N teams for MRM.

Files: `radio/MTP-F26_TeamC_Radio_LinkBudget.xlsx` (hand-in link budget, live formulas) · `radio/linkbudget.py` (same model for quick what-ifs) · `radio/build_workbook.py` · `radio/phy_config.py` (radio profiles and checks) · `radio/fw/` (driver and tests) · `radio/DECISIONS.md`, `HANDOFF_ET.md`, `NM_PHY_PROPOSAL.md`, `TEST_PLAN.md`, `ATENEA_QUESTIONS.md`. Ask Santiago for copies.

Primary sources: competition rules PDF (MTP F26, Network Upgrade III + Sustainability); Session #1 slides; AN5142 with the course annotations; CNAF 2017/2021 (BOE); datasheets for RFM69HCW v1.1, Si4463/61/60-C Rev 1.0, CC1101 SWRS061I, nRF24L01+ v1.0 and CC2500.